Computational Finance — Lecture 1A

Introduction and overview of asset classes

Dr. Fang Fang

TU Delft

- Course nr: WI4151
- Dr. Fang Fang
- Assis. Prof. of Applied Mathematics of TU Delft
- F.Fang@tudelft.nl;

Lecture 1 A - Introduction and overview of asset classes

- General introduction
  - Financial instruments
  - Interest rate definition
  - Financial derivatives
  - Commodities
  - Currencies
  - Cryptos
  - Options

Financial instruments

- Definition:
  - Contracts for monetary assets that can be purchased, traded, created, modified, or settled for. In terms of contracts, there is a contractual obligation between involved parties during a financial instrument transaction.
- Listed products
  - Equities
  - Options
  - Bonds
  - Futures and options on futures
  - Credit Default Swaps (CDSs)
  - Etc.
- Over-the-counter (OTC) products
  - Forwards
  - Swaps and swaptions
  - Options with exotic features
  - Structured products
  - Securitizations
  - Etc.

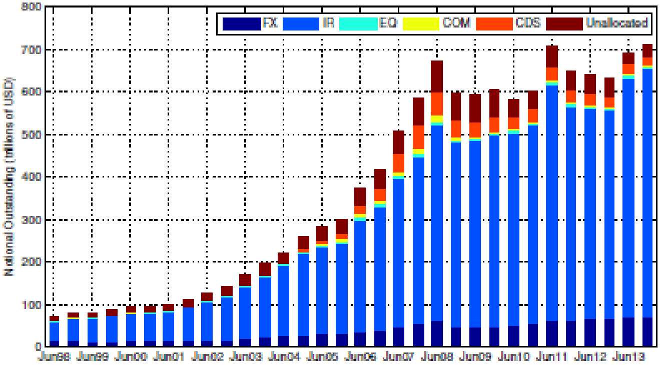
Classification based on asset type

- Equity
- Interest rate (IR) and inflation
- Foreign exchange (FX)
- Commodity & energy
- Credit
- Structured products

Market share

- Equities are only a tiny bit of the market, but the pricing theory of Equity options is fundamental to all products that have optionality.

Market share



Financial instruments - equities/stocks

- A basic financial instrument is equities (stocks, shares)
  - Representing ownership of a tiny part of a company. The price is determined, amongst others, by the present value of the company and the expectation of the company's future performance.
  - These expectations are observed in the bid-ask prices in the market.
  - These expectations give rise to an uncertainty in the price development of a stock in the future. The exact amount of profit is only known at the time the stock is sold.
  - The value of a stock is sometimes somewhat higher, sometimes somewhat lower than its expected value.

Financial markets

- By asset classes
  - Stock market (selling and buying stocks)
  - Interest rates market
  - FX market
  - Credit market
  - Commodity market (gold, metals, corn, meat, futures, . . .)
  - Energy market (oil, gas, . . .)
  - Digital assets such as crypto currencies
- By trading format:
  - Option exchanges (trading nancial options)
  - OTC market (over the counter, not traded on a regulated exchange)

Dividends

- The owner of a stock theoretically owns a piece of the company.
- To the average investor, the value in holding the stock comes from dividends and any growth in the stock's value. Dividends are lump payments, e.g., paid out every six months, to the stock owner.
- The amount of dividend varies from year to year depending on the profitability of the company.
- The amount of dividend is decided by the board of directors and is usually set a month before the dividend is actually paid.
- When the stock is bought, it either comes with its entitlement to the next dividend (cum) or not (ex). There is a date around the time of dividend payment when the stock goes from cum to ex.

The money market

- Interest is money paid for a credit or similar liability. Examples are bond yields, interest paid for bank loans, and returns on savings.
- Interest is typically calculated as a percentage of the principal, the amount owed to the lender. The percentage that is paid over a certain time period is the interest rate. Interest rates are market prices which are determined by supply and demand.
- As money becomes a commodity, the money market became a component of financial markets for assets involved in short-term borrowing, lending, buying and selling.
- Money markets, which provide liquidity for the global financial system, are part of the financial market.
- Trading in money market is done directly with a financial counterparty (over the counter, OTC).
- Interest rates, derivatives and their models are part of the follow up course.

Reasons for interest rate changes

- Banks may change the interest rate to either slow down or speed up economic growth.
- Lowering interest rates can give the economy a short-run boost.
- Generally, if the economy is strong then the interest rates will be high, if the economy is weak the interest rates will be low.
- Economies typically exhibit inflation, a given amount of money buys fewer goods in the future than it will now.
- Risks of investment: the borrower may go bankrupt, or default on the loan. A lender generally charges a risk premium to ensure that she/he is compensated for those that fail.
- Taxes: Because some of the gains from interest may be subject to taxes, the lender may insist on a higher rate to make up for this loss.

Time Value of Money

- The simplest concept in finance is the time value of money.
  - One € today is worth more than one € in a year's time.
  - Invest one € with a discrete interest rate of r (assumed to be constant), paid once per year. After one year your bank account contains
$$1 \times (1 + r).$$
  - There are several types of interest:
    - There is simple and compound interest. Simple interest is when interest you receive is based only on the amount you initially invest; compound interest is when you get interest on your interest.
    - Interest typically comes in two forms, discretely compounded and continuously compounded.

Interest rate – continuous compounding

- With $M(t)$ in the bank at time t, how does this increase with time?
- If you check your account a short period later, time $t + dt$, the amount will have increased by

$$M(t + dt) - M(t) \approx \frac{dM}{dt}\, dt + \cdots \qquad \text{Taylor series expansion}$$

- The interest you receive must be proportional to the amount you have, M, the interest rate r and the time step dt. Thus

$$\frac{dM}{dt}\, dt = rM(t)\, dt \quad \text{giving} \quad \frac{dM}{dt} = rM(t).$$

- If you have $M(0)$ € initially, then solution is

$$M(t) = M(0)e^{rt}.$$

- Conversely, if you know you will get 1 € at time T in the future, its value at an earlier time t is simply

$$e^{-r(T-t)}.$$

Interest rates

OVERVIEW

CURVE CONSTITUENTS

Tenor	Bid	Ask	Net Change	Bid Yield	Ask Yield	Yield Net Change	Aswp Spread	Swap Spread	Z-Spread
1M	100.040	100.041	0.002	-0.684	-0.704	-0.034	-14.1	-15.4	-14.3
3M	100.155	100.177	-0.001	-0.612	-0.699	-0.001	-9.8	-8.2	-9.9
6M	100.336	100.356	0.001	-0.615	-0.652	-0.002	-9.7	-8.5	-10.0
9M	100.382	100.405	-0.006	-0.611	-0.649	0.007	-10.0	-8.1	-10.1
1Y	100.537	100.608	–	-0.611	-0.691	–	-7.8	-8.1	-7.7
2Y	101.356	101.373	–	-0.699	-0.707	–	-17.2	-17.9	-17.2
3Y	107.812	107.842	0.005	-0.740	-0.749	-0.002	-24.6	-23.4	-23.7
4Y	105.140	105.163	0.001	-0.733	-0.739	–	-26.4	-25.7	-26.1
5Y	103.463	103.476	0.001	-0.716	-0.718	–	-26.8	-26.3	-26.6
6Y	105.898	105.931	0.009	-0.695	-0.700	-0.001	-30.4	-29.7	-29.9
7Y	104.563	104.618	-0.031	-0.650	-0.658	0.004	-29.0	-28.5	-28.8
8Y	107.155	107.177	-0.012	-0.610	-0.613	0.001	-31.0	-29.8	-30.4
9Y	105.173	105.193	-0.001	-0.553	-0.555	–	-29.9	-29.1	-29.7
10Y	105.124	105.149	-0.009	-0.494	-0.496	0.001	-28.8	-28.1	-28.6
15Y	104.800	104.933	-0.036	-0.326	-0.335	0.002	-28.9	-28.5	-29.0
20Y	201.167	201.327	-0.132	-0.292	-0.297	0.004	-45.0	-35.3	-30.4
25Y	169.780	169.860	-0.097	-0.167	-0.169	0.003	-20.1	-23.7	-22.0
30Y	102.669	102.778	-0.012	-0.089	-0.093	–	-14.8	-13.9	-15.0

CURVE CHART

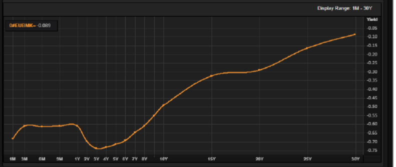


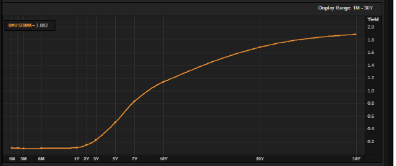
Figure: Interest Rates (Yield Curve) for EUR (sounce: Reuters)

OVERVIEW

CURVE CONSTITUENTS

Tenor	Bid	Ask	Net Change	Bid Yield	Ask Yield	Yield Net Change	Aswp Spread	Swap Spread	Z-Spread
1M	0.095	0.070	0.010	0.096	0.071	0.010	-2.4	-9.7	-2.3
2M	0.095	0.070	0.010	0.096	0.071	0.010	-6.3	–	-6.3
3M	0.085	0.080	–	0.086	0.081	–	-11.3	-10.8	-11.4
6M	0.090	0.085	–	0.091	0.086	–	-8.6	-10.2	-8.7
1Y	0.103	0.095	0.003	0.104	0.096	0.003	-7.8	-9.0	-7.9
2Y	99*30¾	99*31	-0*00½	0.145	0.143	0.008	-6.4	-6.7	-6.4
3Y	99*22¾	99*23	-0*00¼	0.224	0.222	0.003	-5.8	-6.2	-5.9
5Y	99*12¾	99*13	-0*03	0.498	0.496	0.019	-6.5	-6.6	-6.7
7Y	98*19¾	98*20	-0*05¾	0.830	0.829	0.027	-1.0	-0.8	-1.0
10Y	97*19	97*19½	-0*08	1.134	1.132	0.027	0.6	1.3	0.7
20Y	94*29	94*29½	-0*10½	1.678	1.677	0.020	13.1	14.7	14.3
30Y	94*05	94*06	-0*13	1.882	1.880	0.019	24.4	26.6	26.9

CURVE CHART



Volatility

- Volatility is a measure for variation of financial prices over time. The symbol σ is used for volatility, and corresponds to standard deviation, not to be confused with variance σ^2.
- A statistical measure of the tendency of a market or security to rise or fall sharply within a period of time.
- Volatility is typically calculated by using variance of the price or return. A highly volatile market means that prices have huge swings in very short periods of time.
- Volatility tends to have a negative correlation with the return of the asset price.

The volatility index VIX and S&P500

- The S&P500 is an American stock market index based on 500 large companies having stock listed on the NYSE or NASDAQ.

Figure: S&P vs. VIX (source tradingview.com)

Why is the valuation of financial instruments needed

- For MtM reporting purpose:
  - Trading book: MtM accounting; end of day reporting
  - Banking book: accrued accounting; yearly, sometimes quarterly
- For trading purpose:
  - Provide price indication to traders
- For risk management purposes:
  - Sensitivities
  - Risk quantification on portfolio level

How to price a financial instrument? (1/2)

- What/who decides the prices?

How to price a financial instrument? (1/2)

- What/who decides the prices?
  - The market!
- Hence, in case market quotes are available:
  - Reproduce the quoted prices using a pricing model by adjusting model parameters.
  - Then fix the model parameters and shock the risk factors to
    - Calculate sensitivities, e.g. how much the price change if the interest rates go up/down, the FX rate changes, credit spreads change, etc.
    - Quantify the risks on portfolio level

How to price a financial instrument? (2/2)

- In case market quotes are not available:
  - Extract the market information from similar but simpler instruments whose prices are available – i.e. we do calibration. For example,
    - Interest rate curves
    - Volatility surfaces
    - Etc.
  - Insert the results from the above as inputs to the pricing function that prices the instrument under consideration
- Further reading: Options, futures and other derivatives By John Hull

Application of derivatives for hedging purpose

- Suppose you have stocks of a company, and you'd like to have cash in two years (to buy a house).
- You wish at least K Euros for your stocks, but the stocks may drop in the coming years. How to assure K Euros in two years?
- You can buy insurance against falling stock prices.
- This is the standard put option (i.e. the right to sell stock at a future time point).
- The uncertainty is in the stock price.

Application of derivatives for proprietary trading

- Suppose you have confidence in the short-term movements pattern of a stock, and you'd like to bet on it.
- Your analysis, experience and/or additional information indicate that some stock will fall/rise in price in the near future. How to make a profit of your forecast of a falling/rising price?
- You can buy an put/call option, or write a call/put.
- The uncertainty is again in the stock price.

But the main purpose is hedging

- Example: Suppose a portfolio with S (shares) and V_p (puts). If the price of S falls, the value of the portfolio depends on the units of S and V_p.
- With a certain number of units, there can be no movement in the value of the portfolio for a short time period.
- A reduction of risk, for example by combining a number of S and V_p in a portfolio is called hedging.
- What about the writers? A writer of a call expects the value of a share to fall.
- However, writers of options use them as a hedge instrument: an insurance to reduce risk against unexpected movements in the market.

Commodities

- Commodities are raw products such as precious metals, oil, food products, etc.
- The prices of these products are unpredictable but often show seasonal effects. Scarcity of the product results in higher prices.
- Most trading is done on the futures market, making deals to buy or sell the commodity at some time in the future.
- Energy markets are commodity markets that deal specifically with the trade and supply of energy (electricity market, oil, gas).
- Energy markets have been liberalized in some countries; they are regulated by national and international authorities (including liberalized markets) to protect consumer rights and avoid oligopolies. Members of the European Union are required to liberalize their energy markets.

Hedging the risk in commodities

- Consider an energy company, starting energy generation from a wind park at sea (with the tallest wind mills available).
- What are the risks involved, and how can we estimate them?
- (maintenance cost, failure due to too much wind, no wind, ...).
- How to deal with no wind, or too much wind?
- Uncertain wind, uncertain electricity prices
- There is a so-called energy market, where utilities can buy "insurance" (energy derivatives), so that they can buy and sell "power" when the generation is not stable.

Foreign Exchange (FX) rates

- The exchange rate is the rate at which one currency can be exchanged for another: This is the world of foreign exchange or FX. Some currencies are pegged together, others float freely.
- Consistency throughout the FX world: If it is possible to exchange dollars for euros and then euros for yen, this implies a relationship between the dollar/euro, euro/yen and dollar/yen rates.
- If this relation moves out of line, it is possible to make arbitrage profits by exploiting the mispricing.
- Central banks can use interest rates as a tool for manipulating exchange rates.

Hedging the risks in FX rates

- A firm will have some business in America for several years.
- Investment may be in Euros, and payment in the local currency.
- Firm's profit can be influenced negatively by the exchange rate.
- Banks sell insurance against changing FX rates. The option pays out in the best currency each year.
- Uncertain processes are the exchange rates, interest rate, but also the counterparty of the contract may go bankrupt!

FX market and volatility

- FX rate is stochastic and volatile – what is driving this?
- There are options on the FX rate, often used for either speculation or hedging.

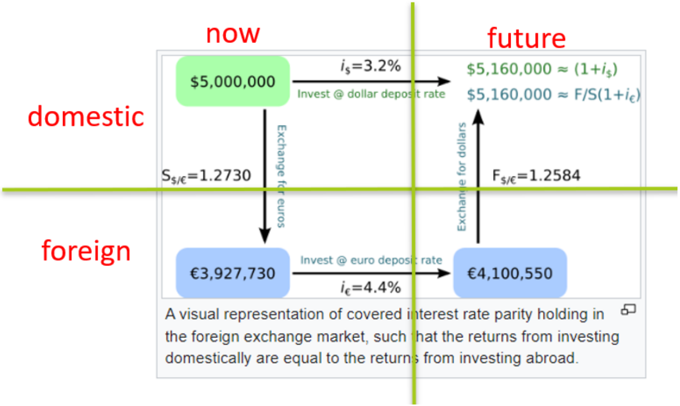
Interest rate parity (1/2)

- Interest rate parity is a no-arbitrage condition representing an equilibrium state under which investors interest rates available on bank deposits in two countries
- Interest rate parity rests on certain assumptions:
  - Capital is mobile - investors can readily exchange domestic assets for foreign assets.
  - Assets have perfect substitutability, following from their similarities in riskiness and liquidity.

Interest rate parity (2/2)

$$(1 + i_s) = \frac{F_t}{S_t}(1 + i_c).$$

- F_t is the forward exchange rate at time t,
- S_t is the spot exchange rate at time t,
- i_s is the foreign (spot) interest rate,
- i_c is the domestic (cash) interest rate.



A visual representation of covered interest rate parity holding in the foreign exchange market, such that the returns from investing domestically are equal to the returns from investing abroad.

In practice: FX basis

- In practice, the interest rate parity does not hold precisely.
- To restore the parity, quants need to add some spread on top of one of the interest rate curves, which is called FX basis.
- This FX basis is treated as information extracted from the FX market and is used back in the pricing of OTC FX derivatives.

Bitcoin

- Crypto currencies behave similarly as regular fiat currencies, but they are much more volatile.
- Bitcoin represents an FX rate of: BTC/USD.

Bitcoin and option market for Cryptos

- Crypto currencies also allow for pricing of options
- Let us take a look a bit at the history of options and types of options available in the market.

Calls					Underlying: SYN.BTC-31DEC21($36127.18)				31 Dec 2021	Expires In 353 days 12 hours 10 minutes								Puts
Last	Size	IV	Bid	Ask	IV	Size	Vol	Δ|Delta	Strike	Last	Size	IV	Bid	Ask	IV	Size	Vol	Δ|Delta
-	-	-	- -	- -	-	-	-	0.94	12000	0.0430	9.8	105.4%	0.0405 $1463.15	0.0525 $1896.68	115.7%	1.0	10.6	-0.06
0.6990	-	-	- -	- -	-	-	-	0.90	16000	0.0725	9.2	104.1%	0.0765 $2766.09	0.0970 $3507.33	116.2%	9.2	12.5	-0.10
0.6335	9.0	89.8%	0.5415 $19638.92	0.5955 $21597.37	115.0%	9.0	-	0.86	20000	0.1385	8.2	104.0%	0.1240 $4484.77	0.1440 $5208.12	113.2%	10.0	24.7	-0.14
0.6030	9.0	91.9%	0.5105 $18267.15	0.5685 $20342.56	116.0%	9.0	0.9	0.84	22000	0.1325	7.6	103.8%	0.1510 $5457.44	0.1740 $6288.71	113.5%	9.0	19.7	-0.16
0.5105	8.2	94.5%	0.4920 $17789.75	0.5430 $19633.81	114.2%	9.0	0.1	0.82	24000	0.1880	7.0	103.8%	0.1800 $6508.45	0.2030 $7340.08	112.7%	8.0	8.5	-0.18
0.5655	9.2	95.5%	0.4670 $16796.61	0.5195 $18684.88	114.4%	9.0	-	0.80	26000	0.2095	6.4	103.8%	0.2110 $7625.96	0.2340 $8457.23	112.1%	10.0	67.4	-0.20
0.4985	8.1	95.1%	0.4400 $15744.46	0.4970 $17784.08	114.4%	9.0	1.4	0.78	28000	0.2445	5.8	103.5%	0.2425 $8768.32	0.2675 $9672.27	112.1%	7.0	42.7	-0.22
0.4500	0.2	95.5%	0.4010 $14504.49	0.4415 $15969.40	108.0%	5.0	7.0	0.74	32000	0.3205	4.7	103.0%	0.3105 $11222.09	0.3335 $12053.36	110.2%	6.0	13.7	-0.26
0.3810	9.0	94.5%	0.3260 $11782.29	0.3750 $13553.25	108.0%	5.0	63.7	0.65	40000	0.4000	4.0	92.0%	0.4235 $15301.39	0.4930 $17812.48	111.1%	5.0	3.6	-0.35
0.3130	8.4	101.9%	0.2995 $10839.27	0.3405 $12323.11	112.7%	9.0	12.0	0.60	48000	0.6800	3.0	0.0%	0.3285 $11868.96	1.0170 $36745.01	223.5%	0.1	4.3	-0.40
0.2630	8.1	103.2%	0.2265 $8189.79	0.2610 $9437.25	112.0%	8.8	17.9	0.50	64000	1.2500	-	-	- -	1.2500 $44719.63	165.7%	1.0	1.0	-0.51
0.2255	8.5	110.8%	0.2030 $7340.08	0.2215 $8009.00	115.6%	8.5	33.6	0.44	80000	-	-	-	- -	- -	-	-	-	-0.57
0.1865	8.9	113.0%	0.1645 $5948.00	0.1835 $6635.00	118.1%	8.9	31.4	0.37	100000	-	0.1	0.0%	0.0005 $17.89	- -	-	-	-	-0.63
0.1450	9.2	114.9%	0.1375 $4967.49	0.1575 $5690.03	120.5%	20.2	33.5	0.32	120000	-	-	-	- -	- -	-	-	-	-0.68
0.1375	1.5	117.1%	0.1200 $4338.96	0.1300 $4700.54	120.1%	10.0	4.0	0.28	140000	-	-	-	- -	- -	-	-	-	-0.72
0.1245	9.6	117.8%	0.1025 $3707.17	0.1220 $4412.44	123.9%	21.6	41.1	0.25	160000	-	0.1	0.0%	0.0005 $17.89	- -	-	-	-	-0.75